

# Supplemental Material for “Geometric Response of Exactly Degenerate Quantum State Bundles”

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(Dated: August 21, 2026)

## I. CUMULANTS, EFFECTIVE CHANNELS, AND THE COMMUTANT TEST

This appendix proves the finite-channel statements used in the main text and fixes the complex covariance convention. None of the arguments assumes a large fiber rank, a thermodynamic limit, or an ETH ansatz.

### A. Realification and weighted cumulants

Let  $\mathcal{V}$  be a finite-dimensional complex vector space and  $\mathcal{V}_{\mathbb{R}}$  its realification. Choose a real basis and denote the real coordinate vector of  $Y \in \mathcal{V}$  by  $y_{\mathbb{R}}$ . For a real dual vector  $t$ , define

$$\Phi_Y(t) = \mathbb{E} \exp(i \langle t, y_{\mathbb{R}} \rangle), \quad K_Y(t) = \log \Phi_Y(t). \quad (1)$$

In a neighborhood of the origin where the required derivatives exist, the order- $k$  cumulant tensor is

$$\kappa_k(Y)[t_1, \dots, t_k] = \frac{1}{i^k} \partial_{s_1} \cdots \partial_{s_k} K_Y \left( \sum_{r=1}^k s_r t_r \right) \Big|_{s_1=\dots=s_k=0}. \quad (2)$$

Finite  $k$ th absolute moments of the tested real-linear projections are sufficient for the derivative statement used here. This construction is real multilinear. Extending it complex linearly recovers cumulants written with  $Y$  and  $Y^\dagger$  in any chosen order.

Suppose  $Y_1, \dots, Y_n$  are mutually independent and take values in the same  $\mathcal{V}$ . A fixed outcome-independent linear identification may be used if the original channel spaces differ. For deterministic real weights and  $Z = \sum_{\alpha} w_{\alpha} Y_{\alpha}$ , independence gives

$$\begin{aligned} \Phi_Z(t) &= \prod_{\alpha=1}^n \Phi_{Y_{\alpha}}(w_{\alpha} t), \\ K_Z(t) &= \sum_{\alpha=1}^n K_{Y_{\alpha}}(w_{\alpha} t). \end{aligned} \quad (3)$$

Substitution of Eq. (3) into Eq. (2) yields

$$\kappa_k(Z)[t_1, \dots, t_k] = \sum_{\alpha=1}^n w_{\alpha}^k \kappa_k(Y_{\alpha})[t_1, \dots, t_k]. \quad (4)$$

Since Eq. (4) holds for arbitrary test fields, it proves the tensor identity quoted in the main text. Centering is required for the simple second-moment formulas and normalized

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fourth statistics used in the paper; the abstract additivity of cumulants itself also holds for noncentered variables when their first cumulants are retained.

Mutual independence is essential. Pairwise independence suffices only up to orders controlled by the corresponding joint laws, and zero cross covariance does not remove a mixed fourth cumulant. If the channel weights depend on the observed  $Y_\alpha$ , Eq. (3) likewise does not apply.

## B. Complete complex Wick subtraction

For a centered complex vector  $z$ , our conventions are

$$C_{ij} = \mathbb{E}[z_i z_j^*], \quad \Pi_{ij} = \mathbb{E}[z_i z_j]. \quad (5)$$

The covariance is Hermitian positive semidefinite and the pseudocovariance is complex symmetric. In terms of  $z = u + iv$ , the pair  $(C, \Pi)$  contains the same information as the full real covariance of  $(u, v)$ . In particular,  $\Pi = 0$  is a property of a proper complex population, not a convention that can be imposed after sampling.

For the ordering  $z_i z_j^* z_k z_l^*$ , Isserlis contraction of a centered complex Gaussian gives

$$W_{i\bar{j}k\bar{l}} = C_{ij}C_{kl} + C_{il}C_{kj} + \Pi_{ik}\Pi_{jl}^*. \quad (6)$$

The corresponding connected tensor is

$$\Gamma_{i\bar{j}k\bar{l}} = \mathbb{E}[z_i z_j^* z_k z_l^*] - W_{i\bar{j}k\bar{l}}. \quad (7)$$

Other index orderings follow from the same realified covariance and contain the analogous three pairings. The scalar contraction used by the executable oracle is

$$\begin{aligned} \hat{\kappa}_4^{\text{conn}} &= \mathbb{E}|z|^4 - 2(\mathbb{E}|z|^2)^2 - |\mathbb{E}z^2|^2, \\ \hat{R}_4^{\text{full}} &= \frac{\hat{\kappa}_4^{\text{conn}}}{(\mathbb{E}|z|^2)^2}. \end{aligned} \quad (8)$$

Equation (8) displays explicitly why ordinary covariance alone is not a complete Gaussian null.

For independent centered channel vectors  $y_\alpha$ , direct expansion gives

$$C_Z = \sum_{\alpha} w_{\alpha}^2 C_{\alpha}, \quad \Pi_Z = \sum_{\alpha} w_{\alpha}^2 \Pi_{\alpha}. \quad (9)$$

At fourth order, all terms involving two distinct independent channels are exactly the Wick pairings constructed from Eq. (9). They cancel in Eq. (7), leaving only  $\sum_{\alpha} w_{\alpha}^4 \Gamma_{\alpha}$ , as required by the general cumulant proof.

### C. Participation number and covariance eigenvalues

Put

$$q_{\alpha} = \frac{w_{\alpha}^2}{\sum_{\beta} w_{\beta}^2}, \quad \sum_{\alpha} q_{\alpha} = 1. \quad (10)$$

Then

$$N_{\text{eff}} = \frac{1}{\sum_{\alpha} q_{\alpha}^2}. \quad (11)$$

This is the inverse participation ratio of the channel powers, not a fitted sample size. If the independent, variance-normalized channels define orthogonal directions in a channel-label covariance matrix  $G = \text{diag}(q_1, \dots, q_n)$ , then

$$N_{\text{eff}} = \frac{(\text{Tr } G)^2}{\text{Tr } G^2} = \frac{(\sum_r g_r)^2}{\sum_r g_r^2}, \quad (12)$$

where  $g_r$  are the eigenvalues of  $G$ . This eigenvalue form explains the usual covariance participation ratio.

Equation (12) must not be applied blindly to the covariance of the flattened physical response. Overlapping channel tensors can have a response covariance whose eigenvalues do not identify the independent units. For correlated channel labels one can define the descriptive quantity  $(\text{Tr } R)^2 / \text{Tr } R^2$  from a label covariance  $R$ , but the mixed connected tensors omitted from Eq. (4) remain. Thus a covariance participation ratio does not restore the theorem when independence is absent.

If every channel has the same variance-normalized connected tensor  $K_4$ , Eq. (4) and Eq. (11) give

$$\frac{\kappa_4(Z)}{(\sum_{\alpha} w_{\alpha}^2)^2} = K_4 \sum_{\alpha} q_{\alpha}^2 = \frac{K_4}{N_{\text{eff}}}. \quad (13)$$

If only  $\|\kappa_4(Y_{\alpha})\| \leq K$  is known, the triangle inequality gives the corresponding  $K/N_{\text{eff}}$  bound. If the tensors are known but unequal, they must remain inside the weighted sum; replacing them by one average is an additional modeling assumption.

#### D. Commutant as a linear system

Let  $A_1, \dots, A_m$  be generators of  $\mathcal{A}_X$ , for example the products  $X_a^\dagger X_b$  together with the adjoints needed for star closure. A matrix  $M$  lies in the commutant precisely when

$$MA_r - A_r M = 0, \quad r = 1, \dots, m. \quad (14)$$

With column-major vectorization,

$$\text{vec}(MA_r - A_r M) = (A_r^T \otimes I - I \otimes A_r) \text{vec } M. \quad (15)$$

Stacking the superoperators in Eq. (15) gives a finite linear system. Its complex nullity is  $\dim \mathcal{A}'_X$ . The executable audit evaluates that nullity by a singular-value decomposition at a stated tolerance. For two Pauli generators it returns one; for one nondegenerate diagonal generator it returns two; for the identity alone it returns four.

For completeness, suppose  $\mathcal{A}_X$  is a finite-dimensional complex star algebra and has a nontrivial invariant subspace. Star closure makes its orthogonal complement invariant as well, and the orthogonal projection onto the subspace is a nonscalar member of  $\mathcal{A}'_X$ . Hence a scalar commutant implies irreducibility. Burnside's theorem then identifies an irreducible unital complex matrix algebra with the full endomorphism algebra. This argument is algebraic. It contains no distribution, cumulant, or large-size statement and therefore supplies no implication from irreducibility to ETH.

#### E. Executable oracle and claim boundary

The synthetic channel population uses independent entries

$$y = \frac{B\xi}{\sqrt{p}}, \quad \Pr(B = 1) = p = 0.1, \quad \xi \in \{1, i, -1, -i\} \quad (16)$$

with the four phases equally likely. It has  $\mathbb{E}y = 0$ ,  $\mathbb{E}|y|^2 = 1$ ,  $\mathbb{E}y^2 = 0$ , and

$$\kappa_4(y) = \mathbb{E}|y|^4 - 2 = \frac{1}{p} - 2 = 8. \quad (17)$$

The audit draws 50,000 complex  $2 \times 2$  matrices, uses 25 blocks for uncertainty, and evaluates equal and linearly unequal weights for  $n = 4, 8, 16, 32, 64$ . Its maximum relative discrepancy from the fixed Eq. (13) prediction is 0.04434; the registered implementation threshold is 0.08. The exponent is not fitted.

All pass/fail decisions use the unrounded double-precision calculation. A recursive ten-significant-digit quantization is applied only when serializing the JSON record and computing its portable scientific hash; nonfinite values are rejected. This serialization convention has no effect on the numerical gate.

The derivation fails or changes in any of the following cases: mutually correlated channels, absent fourth moments, outcome-dependent weights, unresolved symmetry blocks, a size-dependent tangent population, a dominant weight that prevents  $N_{\text{eff}}$  from growing, or a closing external gap. These are hypotheses to test, not technical qualifications that may be discarded after observing a mismatch. In the prospective chiral calculation the fixed random-channel validation fails, so the synthetic oracle and exact proof cannot be promoted to a physical Geometric-ETH conclusion.

## II. CHIRAL-INDEX CONSTRUCTION AND SEALED PROTOCOL

For a full-row-rank factor  $T$ , a full singular-value decomposition  $T = U\Sigma V^\dagger$  supplies the kernel frame  $K = V_{[:,n_b]}$ . Production responses are evaluated by solving

$$(TT^\dagger)Y_a = \dot{T}_a K, \quad \dot{K}_a = -T^\dagger Y_a, \quad (18)$$

for all 16 tangents in one block. We do not diagonalize the dense chiral Hamiltonian in production. Dense diagonalization is retained as an independent small-size oracle for the index and gap.

At every base, the first tangent is also checked through the centered difference

$$\frac{P(T + \epsilon \dot{T}) - P(T - \epsilon \dot{T})}{2\epsilon}, \quad \epsilon = 2 \times 10^{-6}, \quad (19)$$

against the analytic derivative. Gauge covariance is tested after independent unitary changes of basis on the two sublattices. Aggregation is refused unless every base passes the rank, gap, kernel, orthonormality, response-equation, horizontal-gauge, finite-difference, gauge-covariance, tangent-normalization, and whitening gates.

The development and validation grids are disjoint in matrix dimensions and seed streams. The prediction file records the source hashes of the theory audit, parent, runner, and sealing script. The validation runner checks those hashes and the prediction sidecar before producing any outcome tensor. It refuses an existing outcome path and requires the outcome timestamp to be later than the seal timestamp.

The base matrix is the uncertainty unit. Each base contributes the mean of its 16 tangent values. The reported interval resamples 64 base means with replacement for 10,000 replicates. The two-sided percentile bounds use Bonferroni quantiles  $0.05/(2 \times 3)$  and  $1 - 0.05/(2 \times 3)$  within each tangent class across the three validation sizes. These intervals were registered for reporting and control separation; the random primary decision uses only whether each point estimate lies in the frozen development band.

The complete calculation contains three validation dimensions, three tangent classes, 64 bases per dimension and class, and 16 tangents per base. It produces 576 independently gated base records and 9,216 tangent observations. No exponent, band endpoint, covariance threshold, control rule, or branch precedence is fitted after validation.

### III. SCOPE OF THE GRAVITATIONAL COMPARISON

The source audit contains 49 records with unique DOI or arXiv identifiers. Its purpose is bibliographic and logical: it states the narrow claim for which each paper is used and does not treat citations as additional numerical observations. The five statements about D1/D5 half-BPS horizonless states, quarter-BPS  $\mathcal{N} = 4$  super-Yang–Mills states,  $\mathcal{N} = 2$  super-JT gravity,  $\mathcal{N} = 2$  SYK Berry curvature, and large Chern numbers are all traced to the direct calculation of Ref. [1]. We do not multiply the apparent evidence by citing papers that only supply surrounding SYK, JT, or AdS/CFT background.

The named “It from ETH” connection refers to Ref. [2], authored by Hao Geng, Ling-Yan Hung, and Yikun Jiang. The name Lingxin Kong does not occur in that primary record and was not uniquely matched to a relevant source. It is therefore excluded from the author attribution and from manuscript claims.

None of the Paper-I condensed-matter models enters the effective-channel plot as an archival point. Their stored registry does not contain response covariances indexed by a documented independent Hamiltonian or disorder unit, so an  $N_{\text{eff}}$  value would require an imputation or a change of exchangeability unit. The retrospective comparison is restricted

to mechanism and claim boundaries and does not enter the Paper-II title gate.

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- [1] Y. Chen, S. Colin-Ellerin, O. Mamroud, and K. Papadodimas, [Chaos of berry curvature for bps microstates](#) (2026), [arXiv:2604.23287 \[hep-th\]](#).
- [2] H. Geng, L.-Y. Hung, and Y. Jiang, [It from eth: Multi-interval entanglement and replica wormholes from large- \$c\$  bcft ensemble](#) (2025), [arXiv:2505.20385 \[hep-th\]](#).